

Length Contraction, the Pole-and-Shed Paradox, and Cosmic-Ray Muons

PHYS 2203 — Introductory Physics III

September 23, 2026

Overview

Last lecture the postulates of special relativity gave us the relativity of simultaneity and time dilation. This lecture develops the third kinematic consequence, **length contraction**, as the companion to time dilation. We then put both effects to work on the classic *pole-and-shed (barn-pole) paradox*, whose apparent contradiction is dissolved by the relativity of simultaneity. Finally we test relativity against experiment: cosmic-ray *muons*, which classically should never reach the ground, are detected at sea level in abundance — exactly as time dilation and length contraction predict.

Recall from last lecture. The time interval between two events depends on the observer:

$$\Delta t = \gamma \Delta t_0, \quad \gamma \equiv \frac{1}{\sqrt{1 - v^2/c^2}}, \quad (1)$$

where Δt_0 is the **proper time**, measured by an observer for whom the two events happen at the same place, and Δt is the dilated time measured by an observer moving at speed v . We use this result to derive length contraction below.

1 Length Contraction

To uncover the companion effect, turn the light clock sideways so that the light travels parallel to the direction of motion. Choose the clock to be at rest in the O' frame, which moves at speed v along the x -axis relative to the O frame.

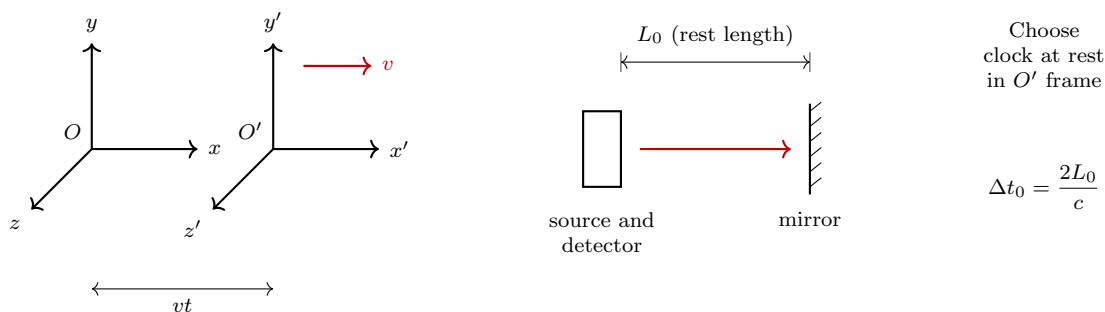


Figure 1: The light clock lying on its side, at rest in the moving frame O' . In its own frame the round-trip time is $\Delta t_0 = 2L_0/c$, where L_0 is the rest length of the clock.

In the O' frame (the clock's rest frame) the light travels a distance L_0 to the mirror and L_0 back, so the proper round-trip time is

$$\Delta t_0 = \frac{2L_0}{c}. \quad (2)$$

The clock as seen in the O frame

In the O frame the clock is moving at speed v . While the light is in flight the mirror moves, so the outbound and return legs cover different distances. Let L be the length of the clock as measured in the O frame.

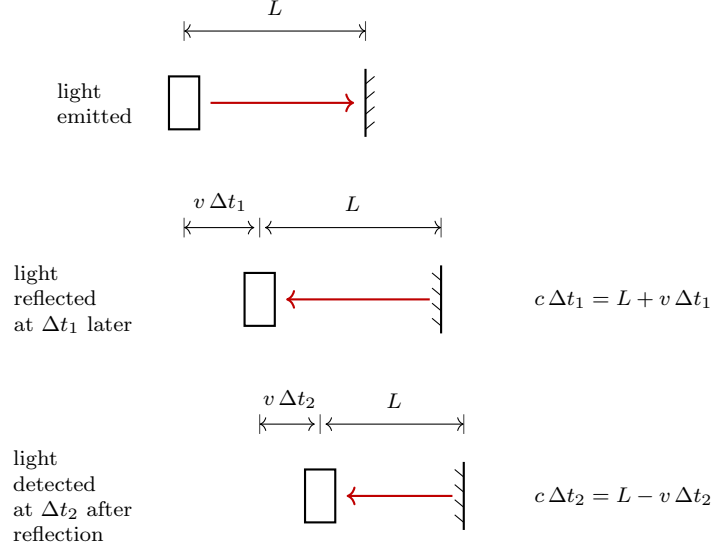


Figure 2: The light clock as seen in the O frame, where it moves to the right at speed v . The light chases the receding mirror on the way out and meets the approaching detector on the way back, so the two legs differ in length.

Outbound leg. The light chases the receding mirror. In time Δt_1 the light travels $c \Delta t_1$ while the mirror moves $v \Delta t_1$:

$$c \Delta t_1 = L + v \Delta t_1 \implies \Delta t_1 = \frac{L}{c - v}. \quad (3)$$

Return leg. On the way back the detector advances to meet the light. In time Δt_2 the light travels $c \Delta t_2$ while the detector moves $v \Delta t_2$ toward it:

$$c \Delta t_2 = L - v \Delta t_2 \implies \Delta t_2 = \frac{L}{c + v}. \quad (4)$$

Round-trip time and length contraction

The total round-trip time in the O frame is

$$\Delta t = \Delta t_1 + \Delta t_2 = \frac{L}{c - v} + \frac{L}{c + v} = \frac{2Lc}{c^2 - v^2} = \frac{2L}{c} \frac{1}{1 - v^2/c^2}. \quad (A)$$

According to time dilation, this same interval is related to the proper time by

$$\Delta t = \gamma \Delta t_0 = \gamma \frac{2L_0}{c}. \quad (\text{B})$$

Comparing (A) and (B),

$$\frac{2L}{c} \frac{1}{1 - v^2/c^2} = \gamma \frac{2L_0}{c} \implies \boxed{L = \frac{L_0}{\gamma}} \quad \text{“length contraction.”} \quad (5)$$

Here L_0 is the **rest length** (or proper length), measured when the object being measured is at rest with respect to the observer. Since $\gamma \geq 1$, a moving object is measured to be shorter along its direction of motion.

Length contraction. The length of an object measured in a frame through which the object is moving is *shorter* than its length measured in the frame in which the object is at rest.

Interpretation as an interval between events

We can think of L as the distance interval between two events:

- Event 1 is the measurement of the position of the left edge.
- Event 2 is the measurement of the position of the right edge.

Which observer needs to be careful about *when* those two measurements are made? The one for whom the object is moving. In the O frame the ends of the clock are going somewhere, so the two position measurements must be made at the same time in O — otherwise the object shifts between them and the difference is not a length at all. In the object’s own rest frame no such care is needed: the ends are not moving, so L_0 may be measured at leisure.

Measuring a moving length. The contracted length L requires the two endpoint events to be *simultaneous in the frame doing the measuring*. The rest length L_0 carries no such requirement. This is why length contraction and the relativity of simultaneity are inseparable — and it is exactly the loophole that resolves the pole-and-shed paradox below.

A warning for problem solving: in most problems the proper *time* Δt_0 and the rest *length* L_0 are measured in *different* frames. In the rocket example that follows, the stations measure the rest length while the astronaut measures the proper time.

Worked example: rocket between space stations

A rocket moves at a constant speed of $0.6c$ from Space Station A to Space Station B, which are 4 light-years apart (as measured by the stations). How long does the trip take according to (i) the astronaut, and (ii) the people on the space stations?

Station (rest) frame. The stations measure the proper *distance* $L_0 = 4$ ly, so the trip time they record is

$$\Delta t = \frac{L_0}{v} = \frac{4 \text{ ly}}{0.6c} = \frac{4 \text{ yr}}{0.6} \approx 6.67 \text{ yr}. \quad (6)$$

Astronaut frame. For the astronaut the distance is length-contracted:

$$\gamma = \frac{1}{\sqrt{1 - 0.6^2}} = 1.25, \quad L = \frac{L_0}{\gamma} = \frac{4 \text{ ly}}{1.25} = 3.2 \text{ ly}. \quad (7)$$

The astronaut therefore measures the (proper) trip time

$$\Delta t_0 = \frac{L}{v} = \frac{3.2 \text{ ly}}{0.6c} = \frac{3.2 \text{ yr}}{0.6} \approx 5.33 \text{ yr}. \quad (8)$$

Equivalently, $\Delta t_0 = \Delta t / \gamma = 6.67 / 1.25 \approx 5.33 \text{ yr}$, consistent with time dilation.

2 The Pole-and-Shed Paradox

Length contraction and the relativity of simultaneity combine in a famous puzzle. A pole vaulter carries a pole of rest length 4 m and runs toward a shed (barn) of rest length 2 m that has a front door and a back door. The pole is at rest in the vaulter's frame; the shed is at rest in the ground frame. At rest the pole is twice as long as the shed.

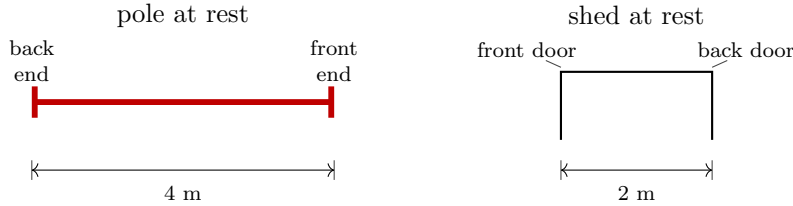


Figure 3: The pole vaulter's 4 m pole and the 2 m shed, each shown in its own rest frame. The pole is twice as long as the shed, so at rest it cannot fit.

How fast to make it fit? In the shed frame the moving pole is length contracted, so it should fit if the vaulter runs fast enough. Setting the contracted length of the pole equal to the shed length,

$$\frac{L_0}{\gamma} = 2 \text{ m} \implies \gamma = \frac{4 \text{ m}}{2 \text{ m}} = 2 \implies v = \frac{\sqrt{3}}{2} c \approx 0.866 c. \quad (9)$$

At this speed $\gamma = 2$, so the moving pole is measured to be $L = 4 \text{ m} / 2 = 2 \text{ m}$, exactly the shed length.

The paradox

Shed frame. The pole is contracted to 2 m and just fits. To test this, an attendant closes the front and back doors *simultaneously* at the instant the pole is fully inside. Both doors shut without striking the pole, then reopen to let it out — the pole *fits*.

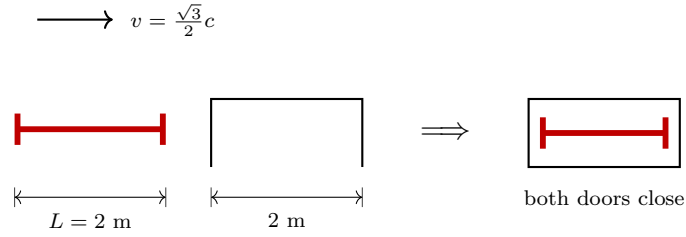


Figure 4: In the shed frame the moving pole is contracted to 2 m, exactly the shed length. Both doors close simultaneously without striking the pole, so the pole *fits*.

Vaulter's frame. Now watch from the pole's rest frame. Here the pole keeps its full 4 m, while the *shed* rushes toward it at $v = \frac{\sqrt{3}}{2}c$ and is contracted to

$$\ell = \frac{2 \text{ m}}{\gamma} = \frac{2 \text{ m}}{2} = 1 \text{ m}. \quad (10)$$

A 4 m pole cannot possibly fit inside a 1 m shed. *This is the paradox:* the shed attendant says the pole fits, the vaulter says it cannot.

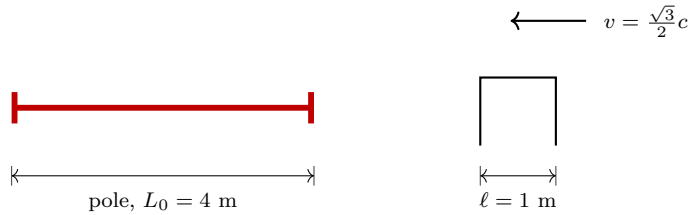


Figure 5: In the vaulter's frame the pole is 4 m and the shed is contracted to 1 m. A 4 m pole cannot fit inside a 1 m shed.

Resolution: the relativity of simultaneity

The escape is the relativity of simultaneity from last lecture: *two events that are simultaneous in one frame need not be simultaneous in another*. The shed attendant's whole test rests on closing the two doors **at the same instant**. But "the two doors close at the same instant" is a statement about two events at *different places*, and the vaulter does not agree that they are simultaneous.

In the vaulter's frame the doors close *one at a time*, not together. The back door (the one the front of the pole reaches first) closes and reopens *before* the front door closes behind the pole's trailing end. In between, the shed slides forward along the pole. There is never a single instant when both doors are shut with the whole 4 m pole inside the 1 m shed. The pole simply passes through.

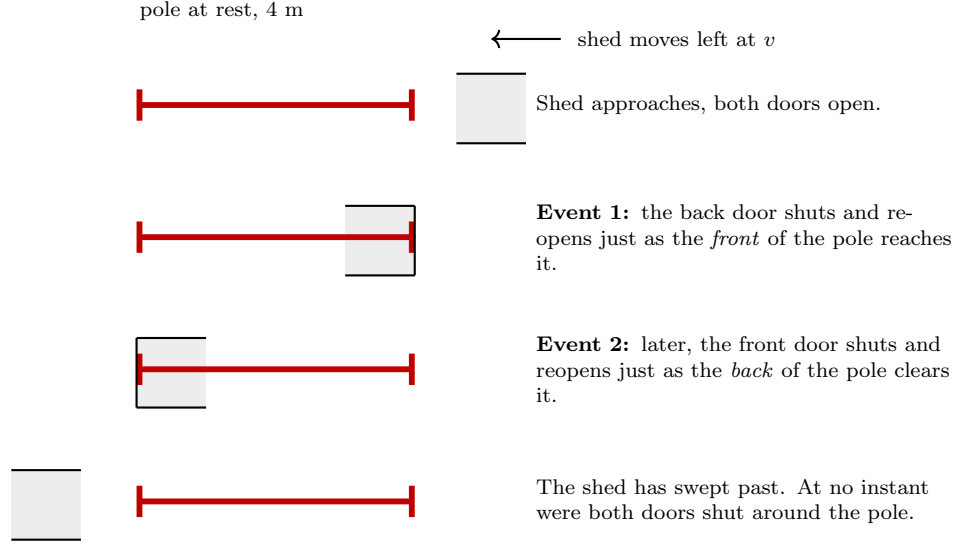


Figure 6: In the vaulter’s frame the doors act one at a time: the back door opens to admit the front of the pole, and only afterward does the front door close behind its trailing end. The pole is never trapped, so both observers agree it is never struck.

Both accounts are self-consistent, and both agree on the only physical fact that matters: the doors never strike the pole. They differ only in the *timing* — simultaneous door closings for the attendant, sequential ones for the vaulter. That difference is precisely the relativity of simultaneity.

What about the pole crashing into the closed back door? One might object that in the shed frame the front of the pole should smash into the closed back door and stop, spoiling the fit. But a signal announcing “the front has stopped” cannot travel back along the pole faster than light. By the time any such signal reaches the trailing end, that end is already inside the shed. Relativity therefore forbids **perfectly rigid objects**: a push or a stop at one end cannot affect the far end instantaneously. Every consistent account is preserved.

3 Experimental Evidence: Cosmic-Ray Muons

Are these effects real? A classic test is provided by **muons**, unstable elementary particles (heavy cousins of the electron) produced when cosmic rays strike the upper atmosphere, roughly 15 km above the ground. A muon at rest decays with a mean lifetime

$$\tau_0 = 2.2 \mu\text{s}. \quad (11)$$

The muons travel at $v \approx 0.9994 c$, for which $\gamma \approx 29$.

The classical puzzle. Without relativity, a muon could travel at most

$$v \tau_0 \approx (3 \times 10^8 \text{ m/s})(2.2 \times 10^{-6} \text{ s}) \approx 660 \text{ m} \quad (12)$$

before decaying — a small fraction of the 15 km it must cross. Essentially none should survive to the ground. Yet detectors at sea level count them in abundance.

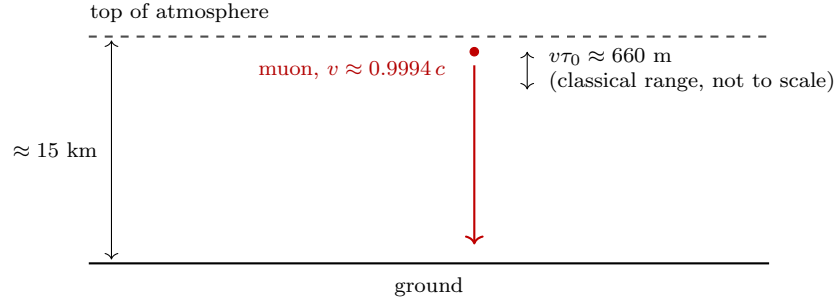


Figure 7: Cosmic-ray muons are created near the top of the atmosphere and detected at the ground — much farther than their classical range $v\tau_0$ would allow.

Ground-frame explanation: time dilation. In the ground frame the muon's internal clock runs slow. Its lifetime is dilated to

$$\gamma \tau_0 \approx 29 (2.2 \mu\text{s}) \approx 64 \mu\text{s}, \quad \text{range} \approx v \gamma \tau_0 \approx 19 \text{ km} > 15 \text{ km}. \checkmark \quad (13)$$

Muon-frame explanation: length contraction. In the muon's own frame its lifetime is just $\tau_0 = 2.2 \mu\text{s}$ — but the atmosphere is moving toward it, and its thickness is length contracted to

$$\frac{15 \text{ km}}{\gamma} \approx \frac{15 \text{ km}}{29} \approx 520 \text{ m}, \quad (14)$$

which the ground covers in $\approx 520 \text{ m} / (0.9994 c) \approx 1.7 \mu\text{s} < \tau_0$. $\checkmark$

The moral. The two frames give different accounts — a dilated lifetime in one, a contracted distance in the other — but agree on the physical fact: *the muons arrive*. Time dilation and length contraction are not optical illusions; particle detectors count their consequences every day. Next lecture we build the **Lorentz transformation**, the single rule relating the positions *and* times of events in two frames, from which time dilation, length contraction, and the relativity of simultaneity all follow at once.